

Environment Setup Guide

(Windows & Mac)

This guide will help you set up your Windows or Mac machine for Java and Spring Boot development. Please follow these steps in order to install and configure all required software.

Prerequisites

- Windows 10/11 or macOS
- Administrator access on your machine
- An active internet connection

Step 1: Install Java Development Kit (JDK) 17

For Windows:

1. Visit the [Oracle JDK 17 Downloads page](#) and download the **x64 Installer** (.exe).
2. Run the downloaded .exe file and follow the default installation instructions.
3. **Set Environment Variables:**
 - Search for "**Edit the system environment variables**" in the Windows Start menu.
 - Click **Environment Variables**.
 - Create a new System Variable named JAVA_HOME and set its value to your JDK path (e.g., C:\Program Files\Java\jdk-17).
 - Find the Path variable, click **Edit**, and add %JAVA_HOME%\bin to the list.
4. Open Command Prompt and type `java -version` to verify.

For Mac:

1. Visit the [Oracle JDK 17 Downloads page](#) and download the **x64 DMG Installer** (for Intel) or **ARM64 DMG Installer** (for Apple M1/M2/M3 chips).
2. Open the .dmg file and follow the installer prompts.

3. Open Terminal and type `java -version` to verify.

Step 2: Install Apache Maven

For Windows:

1. Download the **Binary zip archive** from the [Apache Maven Download page](#).
2. Extract the zip file and move the folder to a permanent location (e.g., `C:\Program Files\maven`).
3. **Set Environment Variables:**
 - Create a new System Variable named `MAVEN_HOME` pointing to your Maven folder.
 - Add `%MAVEN_HOME%\bin` to your Path variable.
4. Open a new Command Prompt and type `mvn -v` to verify.

For Mac (Using Homebrew - Recommended):

1. If you don't have Homebrew, install it by running this in Terminal:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```
2. Install Maven by running: `brew install maven`
3. Verify by running `mvn -v`.

Step 3: Install Visual Studio Code (VS Code)

1. Go to the [VS Code download page](#).
2. **Windows:** Download the Windows installer. Ensure you check the "Add to PATH" option during installation.
3. **Mac:** Download the Mac version, unzip it, and drag the `Visual Studio Code.app` into your **Applications** folder.

Step 4: Install Spring Boot Extension Pack in VS Code

1. Launch VS Code.
2. Click the **Extensions** icon in the left sidebar (or press Ctrl+Shift+X / Cmd+Shift+X).
3. Search for "**Spring Boot Extension Pack**".
4. Click **Install** on the pack published by **VMware**. *Note: We also recommend installing the "Extension Pack for Java" by Microsoft for complete language support.*

Step 5: Install Postman

Postman is used to test your REST APIs.

1. Go to the [Postman Download page](#).
2. **Windows:** Download and run the .exe installer.
3. **Mac:** Download the version for your chip type, unzip it, and move it to the **Applications** folder.

Alternative Tools (Optional)

If you prefer different tools for development or API testing, we recommend the following:

1. IntelliJ IDEA (Alternative to VS Code)

IntelliJ is a powerful IDE designed specifically for Java development.

- **Download:** [JetBrains Website](#)
- **Edition:** **Community Edition** is free and sufficient for most tasks, while **Ultimate Edition** offers advanced Spring Boot features.
- **Setup:** Follow the standard installer for Windows (.exe) or Mac (.dmg).

2. Bruno (Alternative to Postman)

Bruno is a lightweight, fast, and open-source API client that saves your collections locally.

- **Download:** [Bruno Downloads](#)

- **Windows:** Use the .exe installer.
- **Mac:** Use the .dmg installer or run `brew install bruno` in Terminal.

Final Check

Congratulations! Your system is now fully configured for Java and Spring Boot development.
Happy Coding!